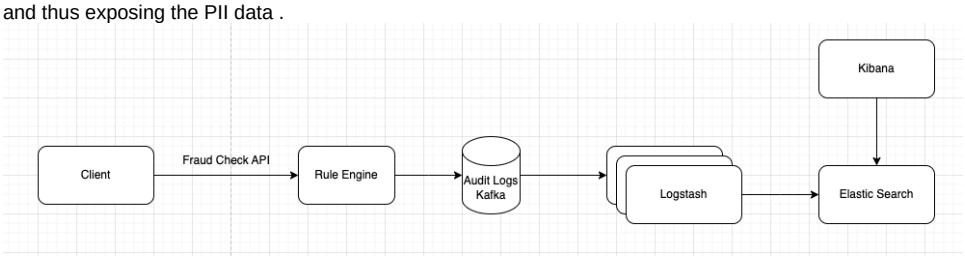
PII Data Masking - POC

- Client calls Maquette fraud check API to execute predefined rules against an input event. The request body of api will have some PII data which is required to validate the input event .
- Maquette Rule Engine will execute the rules and send the audit logs to a Kafka topic. **This audit log also prints the request payload**
- Logstash servers read the audit logs from same kafka topic and send it to elastic search indexes . These logs can then be accessed in Kibana and thus exposing the PII data .



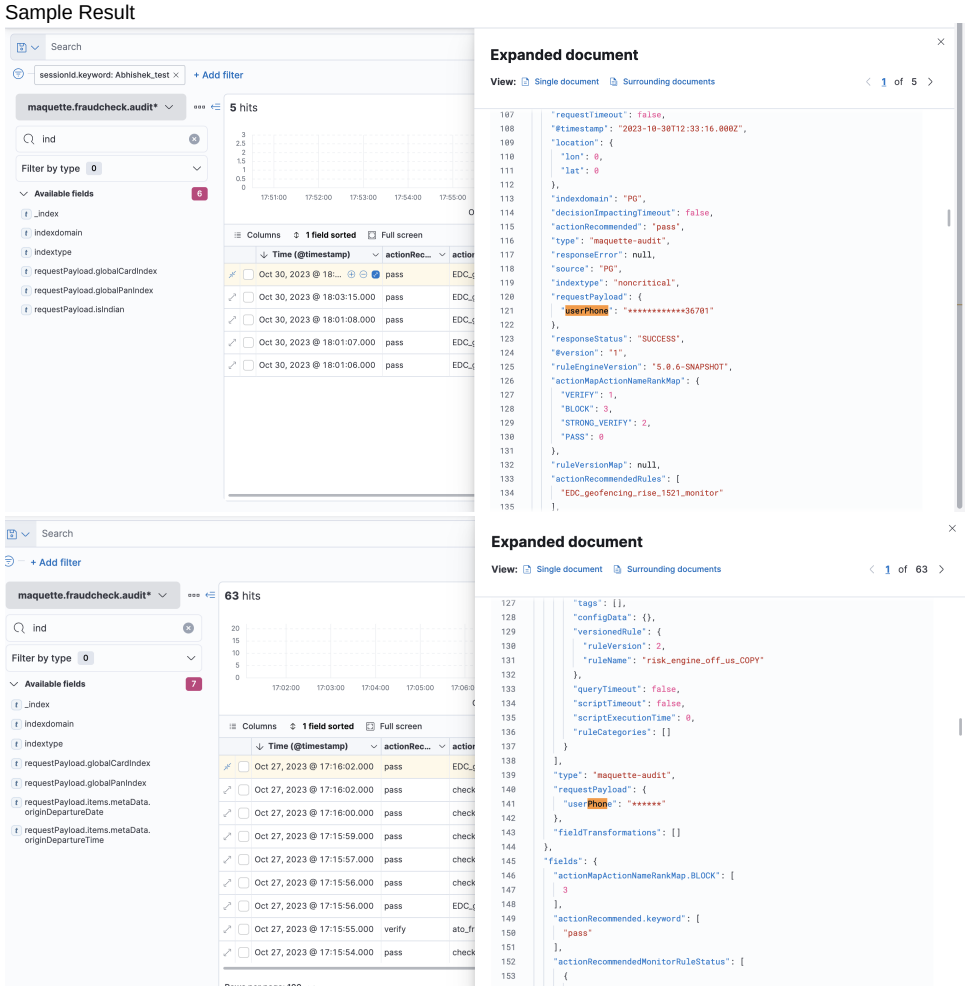
- We can restrict the data that goes into elastic search inside Logstash via logstash filters. We can apply various transformation like replace, add, remove or string substitution to such fields.For this we need to add conf files in logstash.Some examples are shown below
    - Replace - Replace entire field with another string

```
if  [requestPayload][userPhone]  {
      mutate {
          replace  => {
              "[requestPayload][userPhone]" => "****"
          }
      }
  }
```

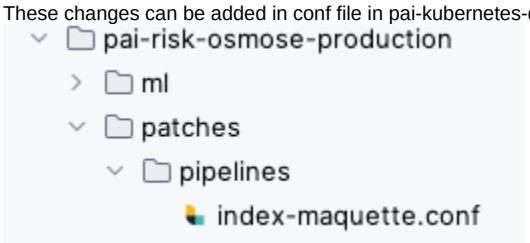
    - Substitute - Replace all except last 4 characters with *

```
if [requestPayload][userPhone] {
      mutate {
      gsub => [ "[requestPayload][userPhone]", "(.*)(\d{4}$)", "************\2"]
      }
  }
```

- Sample Result



- These changes can be added in conf file in pai-kubernetes-deployments repo and it is environment specific.



Reference file for logstash change for M* OCL **pai-risk-maquette/overlays/pai-risk-ocl-production/patches/pipelines/index-maquette.conf**

- Reference file for logstash change for M* Lending **Need to make new file for lending**
- Redeploy logstash server pods in argo cd to see the changes reflected in kibana